
Universal Lies: A multi-method analysis of hallucinations

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Abstract

Hallucinations in large language models remain poorly understood at the mechanistic level. Although prior work has established that LLMs represent the truth and falsehood of statements in their hidden-state activations, these findings rely on simple true/false factual datasets; the mechanistic origins of this separation under naturalistic hallucination-inducing settings, and in particular the role of attention dynamics and individual neurons, have not been directly examined across model families. We address this gap through a multi-method empirical survey spanning five models across four families Qwen2.5, Qwen3, Llama-3.1, and Mistral using TruthfulQA as a common evaluation setting. Our findings include, hallucinating models attending significantly less to question-region tokens in late-layer heads, suggesting a progressive decoupling from the input query before hallucinated content is produced. Hidden-state geometry shows directional separation consistent with prior findings, now extended to naturalistic hallucination-inducing prompts; complementing this, probes trained on hidden states achieve a peak AUC of 0.85 on Qwen2.5-1.5B-Instruct, confirming that a consistent, detectable signature persists across layers. Targeted ablation of hallucination associated neurons disrupts this geometric structure implicating them as a potential contributor to the representational patterns associated with hallucination.

1 Introduction

Hallucination remains a persistent failure mode in large language models, manifesting across domains from question answering to open-ended generation Ji et al. [2023]. Recent work has shown that LLMs linearly represent the truth or falsehood of factual statements in their hidden states Marks and Tegmark [2024]. Since hallucinated responses to factual questions are, at their core, false statements produced by the model, a related geometric structure is observable during generation and that its internal origins can be traced through complementary methods. Rather than proposing a unified theory, we conduct a multi-method empirical survey, probing hallucination from different internal perspectives: attention patterns, hidden state geometry, diagnostic probes Belinkov [2022], Azaria and Mitchell [2023], and targeted neuron ablation Gao et al. [2025]. We apply this framework across five models spanning four families: Qwen2.5-1.5B-Instruct Team [2024], Qwen3-1.7B and Qwen3-4B Team [2025], Llama-3.1-8B-Instruct Grattafiori et al. [2024], and Mistral-v0.2-7B-Instruct Jiang et al. [2023], using TruthfulQA Lin et al. [2021] as a common evaluation setting.

Our findings include, that during hallucination, models attend significantly less to question-region tokens in late-layer heads which is consistent with Chuang et al. [2024] and Liu et al. [2025]. Using per-head consistency analysis to identify the most discriminative head per model, we find that

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hallucinating models redirect attention away from the question context toward induction-only patterns in the final layers of the network. This suggests the model progressively decouples from the input query before producing false content. We additionally find that probes trained on hidden states achieve a mean AUC of 0.80 across model families, with a peak AUC of 0.85 on Qwen2.5 1.5B Instruct confirming a consistent detectable hallucination signal in internal representations. Geometric analysis reveals limited directional separation in hidden states, which we attribute in part to our mean-pooling aggregation strategy collapsing positional information across the token sequence.

Targeted ablation of H neurons Gao et al. [2025] - a sparse set comprising approximately 0.1% of MLP neurons collapses the geometric separation between truthful and hallucinated hidden states across all layers. The layerwise PCA structure that distinguishes the two response types diminishes under ablation, indicating that these neurons actively organize the representational geometry rather than merely correlating with it.

2 Related Work

Hallucination in large language models. Hallucination in LLMs has been studied extensively as a failure mode in open-ended generation and question answering. Early work characterized hallucination taxonomically, distinguishing intrinsic from extrinsic hallucinations Maynez et al. [2020]. More recent work such as Yu et al. [2024], Sun et al. [2024] and Orgad et al. [2024] has shifted the focus towards understanding the internal conditions under which hallucination occurs, motivating mechanistic approaches such as ours.

Attention on question tokens. Prior work has established that truthfulness-relevant signals are unevenly distributed across attention heads Li et al. [2023], and that attention heads and MLPs in upper transformer layers are disproportionately implicated in hallucinations Yu et al. [2024]. Building on this, Ho et al. [2024] showed that attention head norms correlate with truthfulness on TruthfulQA, using them as a lightweight voting signal for hallucination detection. Complementary work has shown that the fraction of attention directed at input tokens serves as a reliable hallucination indicator Chuang et al. [2024], Vazhentsev et al. [2025]. Our experiment extends this line by using key norm differences to *localize* the most discriminative late-layer head per model, and then quantifying how much of that head’s attention budget is devoted to question-region tokens, finding a consistent reduction in question-grounding during hallucinations across model families.

Geometry of internal representations. Marks and Tegmark [2024] demonstrated that truth and false response representations occupy geometrically distinct regions of hidden state space, providing the foundational motivation for our geometric characterization experiments.

Probing classifiers and mechanistic interpretability. Probing classifiers have been widely adopted as a tool for interrogating what information is encoded at each layer of a neural network Belinkov [2022]. Critically, Skean et al. [2025] demonstrate that intermediate layers consistently encode richer representations than final layers across a range of architectures and tasks, directly motivating our per-layer evaluation strategy. Our results (Section 4.3) corroborate this finding in the hallucination setting: peak probe discrimination emerges at 43-58% of network depth across all five models, well before the final layer.

H neurons and structured hallucination circuits. Most directly related to our ablation experiments, Gao et al. [2025] identified a class of neurons whose activations are directly associated with hallucinated outputs. Our work extends this line of inquiry by examining whether ablating these neurons disrupts the geometric separation between truthful and hallucinated hidden states, providing a causal complement to the correlational findings of prior probe-based approaches.

3 Experiments

3.1 Discriminative Heads Attention Analysis via Key Norms

3.1.1 Data Collection

A dataset of 500 questions was drawn from TruthfulQA. Four language models were evaluated: Qwen 2.5 1.5B instruct, Qwen3-1.7B, Qwen3-4B, and LLaMA-3.1-8B-Instruct. For each model - generated answers, Key tensors, Value tensors, and post-softmax attention weights at every other transformer layer were extracted using transformer lensNanda and Bloom [2022]. New token generation was capped at 50 tokens with a sampling temperature of 0.1 to encourage near-deterministic output. Ground-truth labels (truthful / hallucinated) were assigned to each generated response using an LLM-as-judgeZheng et al. [2023] protocol.

3.1.2 Experimental Setup

For each model, per-head key and value norms were computed at answer token positions across all layer-head pairs, yielding matrices of shape $[n_{\text{layers}} \times n_{\text{heads}}]$. Norms were computed as the mean ℓ_2 norm across the token dimension at answer positions only. Question tokens were excluded and then averaged across samples separately for the truthful and hallucinated populations. For grouped-query attention (GQA) Ainslie et al. [2023] models, key and value head counts were derived directly from the stored tensors rather than assumed from the number of query heads, ensuring correct dimensionality handling across architectures.

Difference matrices were computed as $\Delta = \mathbf{M}_{\text{truth}} - \mathbf{M}_{\text{halluc}}$ and visualised as heatmaps. These heatmaps were interpreted by examining where structure appears. If no structure is present and colour is uniform across all heads and layers, this points to a global representational failure with no specific circuit responsible. If certain heads stand out consistently across many layers, a sparse head-level circuit may be implicated. If instead certain layers stand out across all heads, the failure is concentrated at a specific depth. If neither pattern holds and bright spots appear scattered randomly, the head-level signal is too noisy to be informative. Finally, if red activity indicating higher hallucinated norms is concentrated in the late layers, this is interpreted as overcompensation: the model writes with greater magnitude but lower coherence when generating uncertain or false content.

3.1.3 Most Discriminative Head

The most discriminative attention head was identified by isolating the final 30% of layers referred to as the late layers. For each head, the mean absolute difference in key norm between the truthful and hallucinated populations across those layers. The head maximizing this quantity was selected as the most discriminative head. Head selection was performed exclusively on key norms, as keys govern the routing of attention via the $QK^\top$ dot product and therefore capture differences in *where* the model attends rather than *what* it writes; this makes key-based selection the more mechanistically interpretable criterion for identifying heads whose behavior changes with response truthfulness. As head indices are model-specific and do not transfer across architectures, cross-model comparisons were made on functional grounds, characterizing implicated heads by their attention pattern structure and the token types they preferentially attend to.

3.1.4 Attention Patterns in the Most Discriminative Head

For the identified discriminative head, post-softmax attention patterns were visualized across the late layers of each model. For each of three representative examples drawn from both the truthful and hallucinated populations, the attention matrix was sliced at answer destination positions, with a white vertical line demarcating the boundary between question and answer tokens. These plots serve as qualitative illustration of the population-level differences in truths and hallucinations.

To quantify these differences, the mean fraction of non-sink attention directed at question-context tokens was computed per sample across all late layers, referred to as the Q -region fraction:

$$Q = \frac{\sum_{0 < t < q_{\text{len}}} A_t}{\sum_{t > 0} A_t + \varepsilon} \quad (1)$$

The fold-change $\bar{Q}_{\text{truth}}/\bar{Q}_{\text{halluc}}$ was computed at every other late layer to quantify the relative difference between populations. To contextualise this, the mean Q -region fraction was plotted alongside the mean answer-region fraction across late layers separately for each population. A fold-change consistently greater than 1.0 indicates that truthful responses devote a larger share of their attention to question-context tokens than hallucinated responses, suggesting that grounding in the original question is a distinguishing feature of faithful generation at this head.

Compute. All experiments were conducted on a single NVIDIA G4 GPU (through google colab). The workload primarily consists of forward passes for key, value, and attention extraction, totaling approximately 3 GPU-hours. LLM-as-judge evaluation was performed using Ollama’s hosted models, requiring approximately 5 hours of wall-clock time. CPU-based analysis required approximately 10 minutes, with peak memory usage of 10GB RAM.

3.2 Geometric Characterization

This experiment investigates the separability of hallucinations and truthful statements by scaling representations along a dedicated truth-axis. To provide empirical evidence of this dissociation, we train probes on the model’s hidden states, a methodology further elaborated in Section 3.3.

We first constructed a dataset comprising 500 prompts sampled from the TruthfulQA benchmark. We systematically extracted response states and the generated textual output by performing a single forward pass for each prompt. Following extraction, we evaluated the generated outputs to establish ground-truth labels. We employed an LLM-as-a-judge framework utilizing GPT-OSS-120BAgarwal et al. [2025]. For each prompt, the evaluator model was provided with the reference ground-truth and the target model’s generated response. The evaluator assessed the semantic accuracy of the generation, assigning a binary label: $y = 0$ for truthful responses and $y = 1$ for hallucinations.

To map hallucination hidden states onto the truthful axis, the dataset is first partitioned into truthful and hallucinatory responses. Because the generated sequences vary in length, we apply mean pooling across the token dimension of the hidden states, yielding a single, fixed-dimensional representation vector for each response. Next, to align the geometric space with truthful generation, both a standard scaler and a Principal Component Analysis (PCA) model were fitted exclusively on the truthful representations. Finally, hallucinated representations are then normalized using the same scaler and projected onto the truth-derived PCA subspace.

Compute. All experiments were conducted on a single NVIDIA G4 GPU (through google colab). The workload primarily consists of forward passes for hidden states extractions totaling approximately 1.5 GPU-hours. LLM-as-judge evaluation was performed using Ollama’s hosted models, requiring approximately 2 hours of wall-clock time. CPU-based analysis required approximately 10 minutes, with peak memory usage of 40GB RAM.

3.3 Diagnostic Probes

To empirically demonstrate that a separable boundary exists between truthful and hallucinated states, we train a suite of diagnostic probes on hidden states across all the layers. To balance computational overhead with statistical significance across varying model scales, we utilize a tiered sampling strategy. We employ $N = 500$ samples for the Qwen2.5 and Qwen3 families, and $N = 200$ samples for the Mistral and Llama3.1 cohorts. This stratified approach ensures a representative distribution for each architecture while maintaining parity in the evaluation pipeline.

To ensure the robustness of our results, we evaluate all probes using a 3-fold cross-validation (CV) scheme. For each fold, the dataset is partitioned into training and held-out test sets (e.g., approximately 333 training samples per fold for the $N = 500$ cohort). Neural architectures are optimized for 100 epochs using the AdamW Loshchilov and Hutter [2017] optimizer with a learning rate of 1×10^{-3} .

and a binary cross-entropy loss function. Performance is primarily quantified using the Area Under the Curve of the Receiver Operating Characteristic (AUC-ROC), providing a threshold-independent measure of the probe’s ability to discriminate between truthful and hallucinated states.

We employ a dual-approach probing strategy to evaluate the separability of hidden states. The first approach uses a diverse suite of classical machine learning classifiers Pedregosa et al. [2011], while the second leverages specialized neural networks.

We evaluate 16 probes across five families, summarised in Table 4. To prevent data leakage across layers, a freshly initialized, unfitted instance of each probe is instantiated per layer l . All activations $\mathbf{h}_l$ undergo z -score normalization prior to classification; for non-probabilistic models, calibration is applied via Platt scaling [Niculescu-Mizil and Caruana, 2005] with 2-fold CV. The neural suite uses GELU activations [Hendrycks and Gimpel, 2016] and LayerNorm [Ba et al., 2016] where indicated.

Compute. All experiments were conducted on a single NVIDIA T4 GPU (through google colab). The workload primarily consists of forward passes for hidden states extractions totaling approximately 1.5 GPU-hours. LLM-as-judge evaluation was performed using Ollama’s hosted models, requiring approximately 2 hours of wall-clock time. CPU-based analysis required approximately 1 hour with peak memory usage of 40GB RAM.

3.4 H neurons

We systematically kill H neurons during generation to check if it causes the seen geometry of the hallucinated and the truthful responses to collapse. We identify these neurons using the procedure described in Gao et al. [2025]. We then perform a targeted ablation for each identified H-neuron: we set their activations to zero during the forward pass and collect the resulting hidden states and generated outputs. This is done using the same TruthfulQA samples as in the sections above, so that a clear difference between the baseline and the ablated responses can be seen.

3.4.1 Data collection

We identify hallucination-associated neurons through a three-stage pipeline: consistency-filtered generation, token-level contribution extraction, and sparse probing. For each of $N=100$ questions sampled from TruthfulQA, we generate $k=10$ independent responses using nucleus sampling ($T=0.1$, $\text{top-}p=0.95$) and judge each response against the reference answer using the LLM-as-a-Judge framework with GPT-OSS-120B as the judge. We retain only questions where all k responses receive a unanimous label either all truthful ($y=0$) or all hallucinated ($y=1$) discarding ambiguous cases. This consistency filter ensures that the internal activations we analyze correspond to stable behavioral modes rather than borderline generations. For each retained example, we extract the answer span s within the generated response using a prompted extraction step, yielding a token-level mask $\mathbf{m} \in \{0, 1\}^T$ over the input sequence of length T .

3.4.2 Experimental Setup

To quantify each neuron’s contribution to the generated output, we compute *Contribution of Each Token Type* (CETT) features across all layers. For layer ℓ with MLP intermediate activations $\mathbf{a}^{(\ell)} \in \mathbb{R}^{T \times d_{\text{ff}}}$ (captured at the input to the down-projection $W_{\text{down}}^{(\ell)}$), we define the contribution of neuron j at token position t as

$$c_{t,j}^{(\ell)} = a_{t,j}^{(\ell)} \cdot \|W_{\text{down},j}^{(\ell)}\|_2 \quad (2)$$

where $\|W_{\text{down},j}^{(\ell)}\|_2$ is the L_2 norm of the j -th column of the down-projection weight matrix, scaling each neuron’s activation by its downstream influence. We then aggregate contributions separately over answer tokens ($\mathcal{A} = \{t : m_t = 1\}$) and context tokens ($\mathcal{O} = \{t : m_t = 0\}$), yielding per-example feature vectors $\mathbf{c}_{\mathcal{A}}^{(\ell)}, \mathbf{c}_{\mathcal{O}}^{(\ell)} \in \mathbb{R}^{d_{\text{ff}}}$ by averaging over the respective token sets. These are concatenated across all L layers into a single feature vector $\mathbf{x} = [\mathbf{c}_{\mathcal{A}}^{(1)} \parallel \mathbf{c}_{\mathcal{O}}^{(1)} \parallel \dots \parallel \mathbf{c}_{\mathcal{A}}^{(L)} \parallel \mathbf{c}_{\mathcal{O}}^{(L)}] \in \mathbb{R}^{2Ld_{\text{ff}}}$. We construct a binary classification task where hallucinated examples ($y=1$) are answer-token CETT vectors from hallucinated responses, and truthful examples ($y=0$) comprise answer-token CETT from faithful responses together with context-token CETT from both. We fit an L_1 -regularized logistic regression ($C=0.01$) on the standardized features, and define H neurons as those with strictly positive

coefficients in the answer-token partition:

$$\mathcal{H} = \{(\ell, j) : w_{\mathcal{A},j}^{(\ell)} > 0\} \quad (3)$$

where $w_{\mathcal{A},j}^{(\ell)}$ is the learned weight corresponding to neuron j in layer ℓ of the answer-token feature block. Across our models, the resulting $\mathcal{H}$ neuron sets comprise approximately 0.1% of total MLP neurons.

3.4.3 Ablating H-neurons

Having identified the set $\mathcal{H}$ of hallucination-associated neurons, we ask whether these neurons are structurally responsible for the geometric separation observed between truthful and hallucinated hidden states. To test this, we perform a targeted ablation: for each hallucinated example in our dataset, we re-run the model’s forward pass with all $\mathcal{H}$ neurons zeroed out and collect the resulting hidden states.

Concretely, for each layer ℓ , we register a forward hook on the MLP up-projection $W_{\text{up}}^{(\ell)}$. During the forward pass of a hallucinated example, the hook intercepts the intermediate activation $\mathbf{a}^{(\ell)} \in \mathbb{R}^{T \times d_{\text{ff}}}$ and applies a binary mask derived from $\mathcal{H}$:

$$\tilde{a}_{t,j}^{(\ell)} = \begin{cases} 0 & \text{if } (\ell, j) \in \mathcal{H} \\ a_{t,j}^{(\ell)} & \text{otherwise} \end{cases} \quad (4)$$

For truthful examples, no modification is applied their activations pass through unaltered. The modified activation(hallucinations) $\tilde{\mathbf{a}}^{(\ell)}$ propagates through the remainder of the MLP and into the residual stream, affecting all downstream layers. After the clipped forward pass, we extract the mean-pooled hidden state $\bar{\mathbf{h}}^{(\ell)} = \frac{1}{T} \sum_{t=1}^T \mathbf{h}_t^{(\ell)}$ at each layer and project them into two dimensions using PCA for visualization as described in 4.2

Compute. All experiments were conducted on a single NVIDIA A100 GPU (through google colab). The workload primarily consists of forward passes for response generation and neuron extractions totaling approximately 4 GPU-hours. LLM-as-judge evaluation was performed using Ollama’s hosted models, requiring approximately 3hrs hours of wall-clock time. CPU-based analysis required approximately 4 hours, with peak memory usage of 20GB RAM.

4 Results

4.1 Discriminative Attention Head Analysis

Across all four models, we find that hallucinated responses are associated with markedly reduced attention to question tokens relative to answer tokens in the most discriminative head per model. The effect is most visually pronounced in Qwen3-4B (Head 2), as shown in Figure 1 with the statistical validation in Table 1

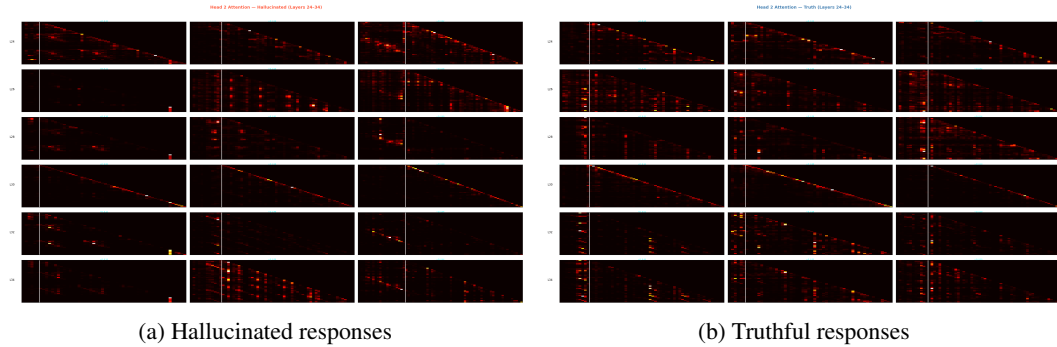


Figure 1: Attention patterns for Head 2 of Qwen3-4B across randomly sampled questions (columns) and late layers (rows). The vertical white line separates question tokens (left) from answer tokens (right). Hallucinated responses exhibit diffuse attention concentrated on answer tokens, whereas truthful responses show substantially higher attention mass on the question region.

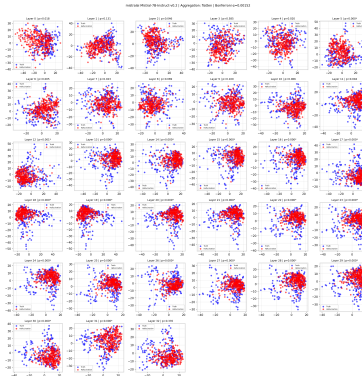
Table 1: Attention fraction analysis for the most discriminative head per model. Layers are sampled from mid-to-late network depth. p -values are from the Mann-Whitney U test; fold-change = $Q\text{-fraction}_{\text{truth}} / Q\text{-fraction}_{\text{halluc}}$. Bold indicates $p < 0.01$. (Mistral-7B-Instruct-v0.2 is excluded from this analysis due to incompatibility with the attention extraction pipeline; all other models are included).

Model	Head	Layer	MWU p (Mann-Whitney U test)	Fold-change
Llama-3.1-8B	H3	22	0.480	1.037
		24	0.210	1.022
		26	< 0.001	1.257
		28	0.035	0.932
		30	0.012	1.095
Qwen2.5-1.5B Instruct	H0	18	0.125	1.046
		20	0.271	1.037
		22	0.325	1.047
		24	0.212	1.056
		26	< 0.001	1.091
Qwen3-4B	H2	24	0.049	0.909
		26	< 0.001	1.088
		28	0.041	1.004
		30	< 0.001	1.074
		32	0.007	1.032
		34	0.110	1.009
Qwen3-1.7B	H6	18	< 0.001	1.086
		20	0.012	1.064
		22	< 0.001	1.099
		24	< 0.001	1.106
		26	0.479	1.024

Table 1 reports the fold-change in question-region attention fraction between truthful and hallucinated responses for the most discriminative head per model, quantifying how much more attention truthful responses allocate to question-context tokens relative to hallucinated ones.

4.2 Geometric Characterization

When we plot the hallucination states with respect to an axis aligned to the truthful states the hallucinated states seem to cluster together as seen in the figure below. The best visually distinguishable example can be seen for Mistral-7B-Instruct-v0.2 as shown below



(a) Mistral-7B-Instruct-v0.2

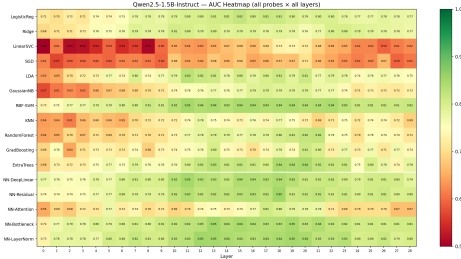
The plots for rest of the models can be found in C

4.3 Diagnostic Probes

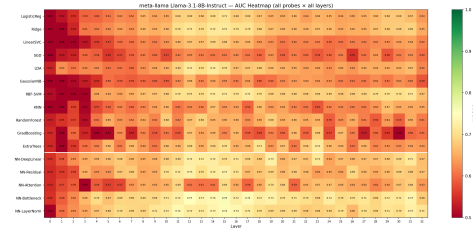
Table 2 summarizes linear and nonlinear probe performance across all five models under evaluation (see Section 3.3). For each model, we report the total layer count, the depth at which peak AUC-ROC is achieved expressed both as an absolute layer index and as a fraction of total network depth, the peak AUC-ROC with cross-validation standard deviation, and the AUC-ROC at layer 0 as a baseline for early-layer discrimination. Across all models, peak discrimination consistently emerges in the middle-to-late layers (43–58% of network depth), with nonlinear probe families dominating throughout. Notably, Llama-3.1-8B exhibits substantially weaker separation across all layers, with a peak AUC-ROC of 0.756 compared to 0.832–0.852 for all other models. Furthermore, layer-0 AUC-ROC remains near chance for Llama-3.1-8B (0.552), whereas the remaining models begin with considerably higher baseline discrimination (0.744–0.774), suggesting that hallucination-predictive geometry emerges earlier and more robustly in those architectures.

Table 2: Cross-model probe summary for hallucination detection on TruthfulQA.

Model	Layers	Peak Layer (%)	Peak AUC-ROC	Dom. Probe	L0 AUC
Qwen2.5-1.5B	29	12 (43%)	0.852 ± 0.011	NN-LayerNorm	0.774
Qwen3-1.7B	29	16 (57%)	0.835 ± 0.015	NN-Residual	0.758
Qwen3-4B	37	21 (58%)	0.847 ± 0.054	NN-Bottleneck	0.762
Mistral-7B	33	15 (47%)	0.832 ± 0.004	NN-Bottleneck	0.744
Llama-3.1-8B	33	15 (47%)	0.756 ± 0.057	NN-Bottleneck	0.552



(a) Qwen2.5-1.5B Instruct



(b) Llama-3.1-8B Instruct

Figure 3: AUC-ROC heatmaps (probe $\times$ layer) for the strongest and weakest performing models. Qwen2.5-1.5B exhibits broad high-discrimination regions from mid-layers onward, while Llama-3.1-8B remains in consistently lower AUC-ROC territory throughout, suggesting that hallucination-predictive geometry emerges earlier and more robustly in the Qwen architecture.

Full per-layer results for all models are provided in AppendixB.

4.4 H Neurons

Figure 4 shows the per-layer PCA projections of hidden states after H neuron suppression on Qwen2.5-1.5B-Instruct. Prior to ablation, truthful and hallucinated representations form visually distinguishable clusters across most layers, consistent with the geometric structure reported throughout this paper. Upon zeroing the H-neurons, the previously separable hidden states become significantly less separable.

To quantify this collapse beyond visual inspection, we train the same suite of diagnostic probes described in Section 3.3 on the post-ablation hidden states across all four models. Table 3 reports the best probe AUC before and after ablation at the same layer. The drop is consistent across every model family: ablation degrades probe discriminability in all cases, confirming that H neurons are load-bearing for the representational distinction between truthful and hallucinated states.

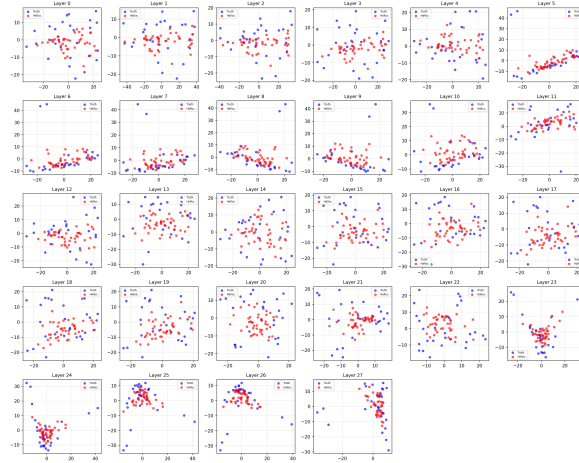


Figure 4: Post ablation Qwen2.5-1.5B

Table 3: Probe AUC before and after H neuron ablation across four models, evaluated at the layer with peak pre-ablation performance.

Model	Best Layer	Pre-Abl. AUC	Post-Abl. AUC
Qwen2.5-1.5B-Instruct	12	0.852	0.690
Qwen3-1.7B	16	0.835	0.540
Qwen3-4B	21	0.847	0.620
Llama-3.1-8B-Instruct	15	0.756	0.580

We emphasise what this result does not show. The ablation collapses the representational distinction; it does not establish that suppressing H-neurons corrects hallucinations at the output level.

5 Conclusion

Hallucinations leave a trace that is structured enough to be locatable and consistent across architectures. Across five models spanning three families, our findings converge on an affirmative answer. When a model hallucinates, it attends significantly less to question-region tokens in late-layer discriminative heads. The answer is already leaving the rails before the first token is generated. Hidden state probes trained on hidden states confirm a detectable signature, peaking at AUC-ROC of 0.852 in the layers just past the middle of the network. Geometric analysis reveals that this signal is present across model families, though its strength varies: Llama-3.1-8B exhibits markedly weaker separation throughout, while Qwen and Mistral architectures develop the signature earlier and more robustly. Finally, targeted ablation of H neurons collapses the geometric separation between truthful and hallucinated hidden states entirely, suggesting the pattern is structurally grounded rather than a statistical artifact. Hallucination leaves a trace inside the model, that trace is structured, locatable, and not accidental.

6 Future Work

Several directions emerge naturally from our findings. First, whether the probes trained in this work can be distilled into lightweight real-time detectors that flag hallucination during generation. Second, our attention head analysis is correlational; future work should apply causal tracing Meng et al. [2022] to the identified discriminative heads to establish whether suppressing question-region attention causally induces hallucination. Third, the geometric separation we observe relies on mean-pooled representations; revisiting this analysis with last-token or weighted aggregation strategies may reveal stronger directional structures.

A Probing Suite

Table 4: Probing suite: classifiers and mechanistic hypotheses tested at each layer l .

Family	Probe	hypothesis tested
<i>Classical suite</i>		
Linear	Logistic Regression	Linear separability of truthful and hallucinated populations
	Ridge Classification	Regularised linear boundary under multicollinearity
	Linear SVC	Maximum-margin linear separation
	SGD (modified Huber)	Robustness to outlier hidden states
Probabilistic	LDA	Gaussian class-conditional geometry
	Gaussian Naive Bayes	Feature independence across hidden dimensions
Kernel / distance	RBF-SVM	Non-linear separability in kernel space
	k -NN ($k=5$)	Local neighbourhood structure of representations
Tree ensembles	Random Forest	Feature importance via bagged decision boundaries
	Extra Trees	Randomised splitting as variance reduction baseline
	Gradient Boosting	Sequential correction of residual errors
<i>Neural suite</i>		
Neural	Deep Linear Probe	Pure linear decomposability via low-rank factorisation
	Residual MLP	Non-linear structure without representational degradation
	Chunked Self-Attention	Feature-level importance across contiguous hidden-state chunks
	Information Bottleneck	Minimum intrinsic dimensionality of the hallucination manifold
	Directional (magnitude-invariant)	Whether angular direction alone discriminates populations

B Full Probe heatmaps and Per layer Probe results

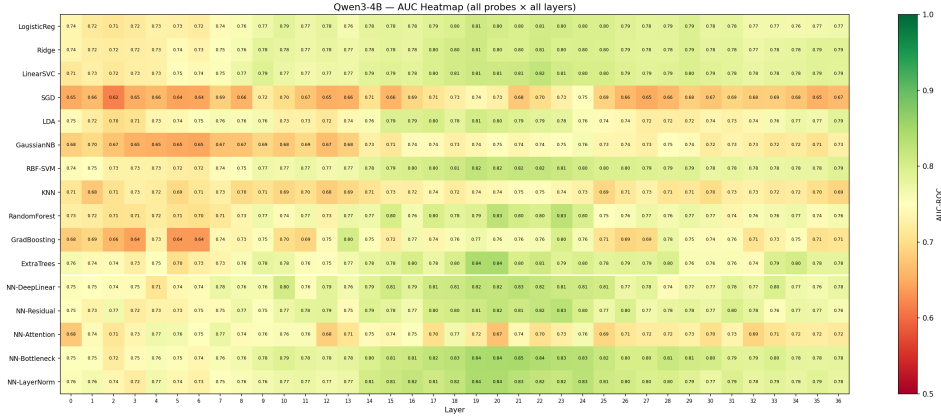


Figure 6: AUC-ROC heatmap (probe × layer) for Qwen3-4B. Peak discrimination occurs later relative to network depth (layer 21, 58%) compared to smaller Qwen variants, with NN-Bottleneck and ExtraTrees achieving the highest AUC-ROC values. The broader high-discrimination plateau spanning layers 19–24 suggests that larger model capacity distributes hallucination geometry across a wider range of layers.

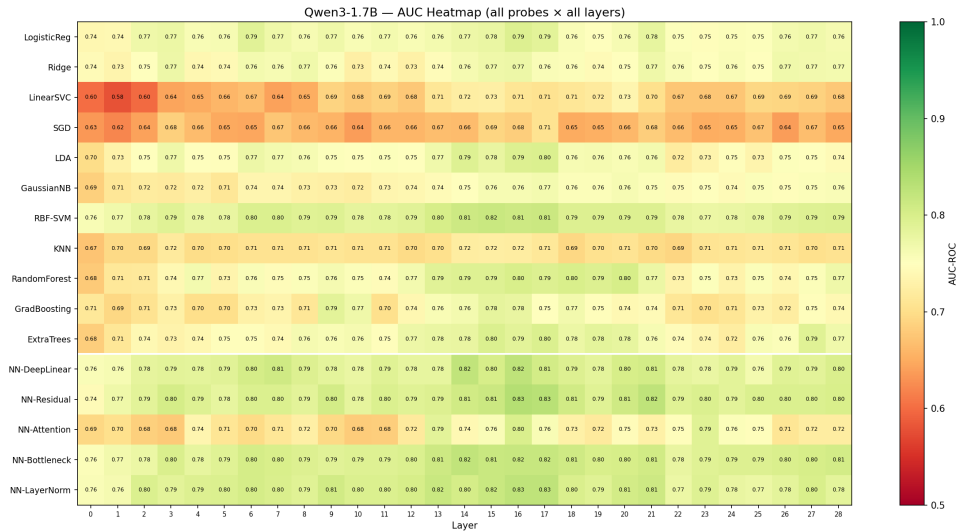


Figure 5: AUC-ROC heatmap (probe × layer) for Qwen3-1.7B. Discrimination builds gradually from layer 0 and peaks around layers 16–18, with NN-Residual and NN-Bottleneck as the dominant probe families. Linear probes such as LinearSVC and SGD remain consistently weaker, indicating that the truthful versus hallucinated separation is not linearly structured in this model’s residual stream.

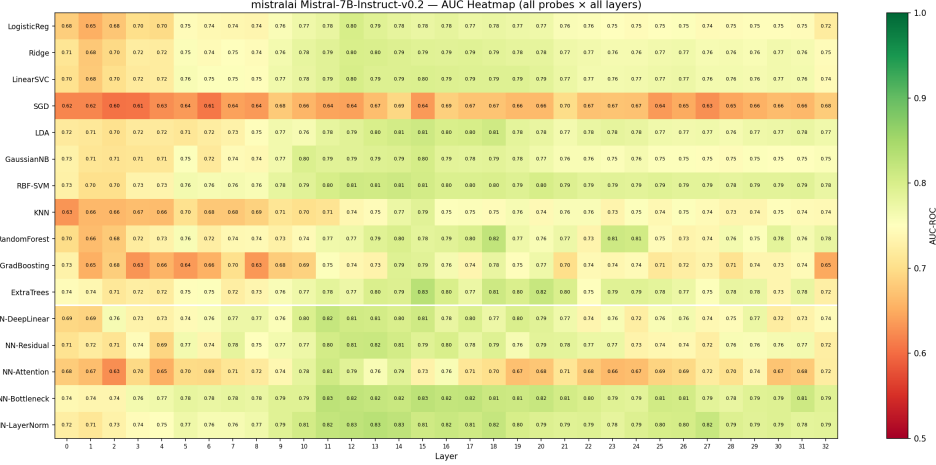


Figure 7: AUC-ROC heatmap (probe × layer) for Mistral-7B-Instruct-v0.2. Discrimination peaks sharply around layers 10–15 before declining slightly in later layers, with NN-Bottleneck and NN-LayerNorm as the strongest probes throughout. SGD remains a consistent outlier with substantially lower AUC-ROC across all layers, consistent with the pattern observed across all models in this study.

Table 5: Best-performing probe per layer for Qwen3-1.7B on TruthfulQA, selected by mean AUC-ROC across cross-validation folds. NN-Residual and NN-LayerNorm dominate throughout, with peak discrimination at layer 16 (57% depth). Discrimination remains strong and stable across the latter half of the network.

Layer	Best Probe	AUC (mean ± std)	AP	Acc	F1
0	RBF-SVM	0.758 ± 0.023	0.880	0.736	0.843
1	NN-Residual	0.771 ± 0.019	0.884	0.730	0.807
2	NN-LayerNorm	0.797 ± 0.017	0.900	0.765	0.843
3	NN-Residual	0.804 ± 0.033	0.910	0.771	0.841
4	NN-LayerNorm	0.793 ± 0.024	0.891	0.738	0.809
5	NN-LayerNorm	0.797 ± 0.024	0.910	0.750	0.830
6	NN-DeepLinear	0.804 ± 0.028	0.909	0.726	0.792
7	NN-DeepLinear	0.810 ± 0.026	0.914	0.726	0.790
8	NN-DeepLinear	0.793 ± 0.025	0.902	0.716	0.777
9	NN-LayerNorm	0.807 ± 0.020	0.910	0.738	0.813
10	NN-LayerNorm	0.796 ± 0.018	0.898	0.751	0.821
11	NN-Residual	0.795 ± 0.013	0.909	0.738	0.812
12	NN-LayerNorm	0.802 ± 0.018	0.901	0.746	0.814
13	NN-LayerNorm	0.815 ± 0.020	0.911	0.756	0.829
14	NN-Bottleneck	0.823 ± 0.023	0.911	0.740	0.804
15	RBF-SVM	0.815 ± 0.019	0.921	0.781	0.863
16	NN-Residual	0.835 ± 0.015	0.930	0.771	0.834
17	NN-Residual	0.832 ± 0.021	0.927	0.757	0.817
18	NN-Bottleneck	0.809 ± 0.032	0.908	0.734	0.803
19	NN-Bottleneck	0.800 ± 0.030	0.906	0.736	0.809
20	NN-Residual	0.810 ± 0.025	0.912	0.752	0.811
21	NN-Residual	0.825 ± 0.010	0.907	0.739	0.800
22	NN-Residual	0.790 ± 0.025	0.897	0.718	0.784
23	NN-Residual	0.796 ± 0.009	0.894	0.724	0.784
24	NN-Bottleneck	0.794 ± 0.022	0.897	0.730	0.798
25	NN-Residual	0.796 ± 0.006	0.898	0.714	0.785
26	NN-Bottleneck	0.798 ± 0.029	0.898	0.726	0.791
27	NN-Bottleneck	0.802 ± 0.027	0.908	0.720	0.787

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Table 5 – continued from previous page

Layer	Best Probe	AUC (mean $\pm$ std)	AP	Acc	F1
28	NN-Bottleneck	0.805 $\pm$ 0.032	0.901	0.714	0.783

Table 6: Best-performing probe per layer for Qwen3-4B on TruthfulQA, selected by mean AUC-ROC across cross-validation folds. NN-Bottleneck dominates from layer 17 onward, with peak discrimination at layer 21 (58% depth). The broader plateau of high AUC-ROC spanning layers 19–24 distinguishes this model from smaller variants.

Layer	Best Probe	AUC (mean $\pm$ std)	AP	Acc	F1
0	ExtraTrees	0.762 $\pm$ 0.037	0.848	0.660	0.765
1	NN-LayerNorm	0.764 $\pm$ 0.033	0.863	0.670	0.713
2	NN-Residual	0.767 $\pm$ 0.019	0.844	0.735	0.779
3	NN-Bottleneck	0.745 $\pm$ 0.040	0.823	0.675	0.725
4	NN-Attention	0.769 $\pm$ 0.049	0.852	0.705	0.739
5	NN-Attention	0.764 $\pm$ 0.024	0.849	0.700	0.739
6	LDA	0.752 $\pm$ 0.041	0.839	0.685	0.756
7	NN-DeepLinear	0.775 $\pm$ 0.031	0.851	0.700	0.741
8	LinearSVC	0.773 $\pm$ 0.039	0.848	0.675	0.784
9	LinearSVC	0.792 $\pm$ 0.035	0.866	0.665	0.778
10	NN-DeepLinear	0.805 $\pm$ 0.018	0.869	0.735	0.779
11	NN-Residual	0.784 $\pm$ 0.094	0.863	0.704	0.749
12	NN-Residual	0.787 $\pm$ 0.041	0.858	0.705	0.758
13	GradBoosting	0.804 $\pm$ 0.049	0.890	0.705	0.787
14	NN-LayerNorm	0.807 $\pm$ 0.078	0.876	0.720	0.773
15	NN-LayerNorm	0.813 $\pm$ 0.061	0.875	0.740	0.779
16	NN-LayerNorm	0.822 $\pm$ 0.050	0.883	0.765	0.796
17	NN-Bottleneck	0.823 $\pm$ 0.058	0.887	0.745	0.789
18	NN-Bottleneck	0.828 $\pm$ 0.054	0.894	0.730	0.776
19	NN-Bottleneck	0.842 $\pm$ 0.049	0.898	0.765	0.808
20	NN-Bottleneck	0.845 $\pm$ 0.044	0.910	0.785	0.823
21	NN-Bottleneck	0.847 $\pm$ 0.054	0.909	0.755	0.785
22	NN-Bottleneck	0.840 $\pm$ 0.048	0.906	0.745	0.786
23	NN-Bottleneck	0.833 $\pm$ 0.046	0.892	0.765	0.803
24	NN-Bottleneck	0.831 $\pm$ 0.060	0.896	0.765	0.801
25	NN-Bottleneck	0.816 $\pm$ 0.055	0.888	0.760	0.800
26	NN-LayerNorm	0.805 $\pm$ 0.051	0.865	0.740	0.788
27	NN-Bottleneck	0.801 $\pm$ 0.045	0.875	0.725	0.770
28	NN-Bottleneck	0.815 $\pm$ 0.042	0.882	0.725	0.765
29	NN-Bottleneck	0.808 $\pm$ 0.052	0.876	0.715	0.750
30	NN-Bottleneck	0.803 $\pm$ 0.054	0.878	0.710	0.762
31	NN-Residual	0.797 $\pm$ 0.037	0.855	0.715	0.741
32	NN-Bottleneck	0.786 $\pm$ 0.040	0.870	0.695	0.742
33	NN-DeepLinear	0.797 $\pm$ 0.043	0.870	0.705	0.744
34	ExtraTrees	0.803 $\pm$ 0.070	0.882	0.725	0.806
35	Ridge	0.789 $\pm$ 0.049	0.877	0.680	0.774
36	RBF-SVM	0.789 $\pm$ 0.063	0.878	0.685	0.782

Table 7: Best-performing probe per layer for Qwen2.5-1.5B-Instruct on TruthfulQA, selected by mean AUC-ROC across cross-validation folds. RBF-SVM and NN-LayerNorm share dominance across mid-to-late layers, with peak discrimination at layer 12 (43% depth) — the earliest peak across all models evaluated. High AUC-ROC is sustained from layer 11 through layer 28.

Layer	Best Probe	AUC (mean $\pm$ std)	AP	Acc	F1
0	NN-DeepLinear	0.774 $\pm$ 0.015	0.941	0.702	0.789
1	NN-LayerNorm	0.784 $\pm$ 0.009	0.938	0.748	0.833
2	NN-Bottleneck	0.786 $\pm$ 0.015	0.941	0.706	0.791
3	NN-LayerNorm	0.782 $\pm$ 0.035	0.933	0.754	0.839
4	NN-Bottleneck	0.796 $\pm$ 0.030	0.942	0.654	0.745
5	NN-LayerNorm	0.804 $\pm$ 0.023	0.941	0.750	0.834
6	NN-LayerNorm	0.800 $\pm$ 0.023	0.942	0.752	0.835
7	NN-LayerNorm	0.817 $\pm$ 0.027	0.950	0.780	0.857
8	RBF-SVM	0.811 $\pm$ 0.017	0.949	0.840	0.912
9	NN-Bottleneck	0.809 $\pm$ 0.012	0.947	0.766	0.838
10	NN-DeepLinear	0.833 $\pm$ 0.034	0.959	0.712	0.797
11	RBF-SVM	0.849 $\pm$ 0.030	0.960	0.854	0.919
12	NN-LayerNorm	0.852 $\pm$ 0.011	0.962	0.772	0.848
13	NN-Bottleneck	0.851 $\pm$ 0.014	0.960	0.808	0.878
14	NN-LayerNorm	0.841 $\pm$ 0.009	0.959	0.802	0.873
15	RBF-SVM	0.845 $\pm$ 0.033	0.957	0.862	0.923
16	RBF-SVM	0.844 $\pm$ 0.030	0.959	0.868	0.926
17	NN-LayerNorm	0.839 $\pm$ 0.012	0.952	0.812	0.882
18	NN-Residual	0.851 $\pm$ 0.026	0.964	0.796	0.868
19	NN-LayerNorm	0.849 $\pm$ 0.017	0.962	0.822	0.892
20	ExtraTrees	0.845 $\pm$ 0.026	0.952	0.850	0.917
21	RBF-SVM	0.840 $\pm$ 0.017	0.961	0.850	0.917
22	RBF-SVM	0.843 $\pm$ 0.011	0.956	0.846	0.915
23	NN-LayerNorm	0.832 $\pm$ 0.006	0.950	0.776	0.852
24	RBF-SVM	0.814 $\pm$ 0.023	0.943	0.838	0.911
25	RBF-SVM	0.815 $\pm$ 0.021	0.943	0.840	0.912
26	NN-LayerNorm	0.815 $\pm$ 0.011	0.944	0.778	0.858
27	RBF-SVM	0.811 $\pm$ 0.017	0.944	0.838	0.911
28	RBF-SVM	0.816 $\pm$ 0.006	0.946	0.840	0.912

Table 8: Best-performing probe per layer for Mistral-7B-Instruct-v0.2 on TruthfulQA, selected by mean AUC-ROC across cross-validation folds. NN-Bottleneck and NN-LayerNorm dominate from layer 9 onward, with peak discrimination at layer 15 (47% depth). AUC-ROC declines gradually in later layers, a pattern distinct from the sustained plateaus observed in the Qwen model family.

Layer	Best Probe	AUC (mean $\pm$ std)	AP	Acc	F1
0	ExtraTrees	0.744 $\pm$ 0.041	0.809	0.715	0.788
1	ExtraTrees	0.738 $\pm$ 0.014	0.807	0.710	0.782
2	NN-DeepLinear	0.760 $\pm$ 0.041	0.810	0.675	0.696
3	NN-Bottleneck	0.765 $\pm$ 0.029	0.825	0.640	0.655
4	NN-Bottleneck	0.768 $\pm$ 0.031	0.834	0.655	0.675
5	NN-Bottleneck	0.780 $\pm$ 0.025	0.849	0.675	0.705
6	NN-Bottleneck	0.776 $\pm$ 0.024	0.828	0.725	0.759
7	NN-Residual	0.783 $\pm$ 0.069	0.835	0.720	0.758
8	NN-Bottleneck	0.775 $\pm$ 0.013	0.846	0.705	0.749
9	NN-LayerNorm	0.790 $\pm$ 0.015	0.857	0.685	0.720
10	NN-LayerNorm	0.805 $\pm$ 0.014	0.870	0.720	0.741
11	NN-Bottleneck	0.827 $\pm$ 0.006	0.886	0.725	0.745
12	NN-LayerNorm	0.826 $\pm$ 0.006	0.885	0.775	0.814

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Table 8 – continued from previous page

Layer	Best Probe	AUC (mean $\pm$ std)	AP	Acc	F1
13	NN-LayerNorm	0.831 $\pm$ 0.022	0.894	0.745	0.777
14	NN-LayerNorm	0.826 $\pm$ 0.029	0.886	0.760	0.791
15	ExtraTrees	0.832 $\pm$ 0.004	0.891	0.735	0.805
16	NN-Bottleneck	0.822 $\pm$ 0.002	0.881	0.755	0.782
17	NN-Bottleneck	0.824 $\pm$ 0.006	0.883	0.750	0.781
18	RandomForest	0.823 $\pm$ 0.022	0.869	0.750	0.813
19	NN-Bottleneck	0.811 $\pm$ 0.012	0.873	0.740	0.766
20	NN-Bottleneck	0.817 $\pm$ 0.010	0.865	0.730	0.765
21	NN-Bottleneck	0.805 $\pm$ 0.024	0.866	0.730	0.759
22	NN-Bottleneck	0.799 $\pm$ 0.016	0.859	0.700	0.727
23	RandomForest	0.812 $\pm$ 0.028	0.867	0.725	0.807
24	RandomForest	0.808 $\pm$ 0.022	0.872	0.720	0.801
25	NN-Bottleneck	0.808 $\pm$ 0.024	0.872	0.700	0.731
26	NN-Bottleneck	0.806 $\pm$ 0.017	0.856	0.740	0.776
27	NN-LayerNorm	0.817 $\pm$ 0.020	0.878	0.735	0.766
28	NN-LayerNorm	0.793 $\pm$ 0.026	0.862	0.740	0.781
29	NN-Bottleneck	0.790 $\pm$ 0.021	0.851	0.715	0.748
30	NN-LayerNorm	0.792 $\pm$ 0.026	0.854	0.665	0.704
31	NN-Bottleneck	0.812 $\pm$ 0.034	0.868	0.705	0.737
32	NN-Bottleneck	0.789 $\pm$ 0.022	0.859	0.670	0.684

Table 9: Best-performing probe per layer for Llama-3.1-8B-Instruct on TruthfulQA, selected by mean AUC-ROC across cross-validation folds. NN-Bottleneck and NN-LayerNorm dominate throughout, with peak discrimination at layer 15 (47% depth). Overall AUC-ROC values are substantially lower than all other models evaluated, with layer-0 discrimination near chance (0.552), indicating that hallucination-predictive geometry develops more weakly across the entire forward pass in this architecture.

Layer	Best Probe	AUC (mean $\pm$ std)	AP	Acc	F1
0	NN-DeepLinear	0.552 $\pm$ 0.042	0.786	0.510	0.604
1	LDA	0.626 $\pm$ 0.028	0.831	0.735	0.842
2	NN-LayerNorm	0.627 $\pm$ 0.054	0.828	0.645	0.759
3	NN-Bottleneck	0.661 $\pm$ 0.053	0.847	0.665	0.763
4	NN-Bottleneck	0.695 $\pm$ 0.034	0.874	0.644	0.720
5	NN-LayerNorm	0.682 $\pm$ 0.050	0.863	0.645	0.745
6	NN-LayerNorm	0.697 $\pm$ 0.053	0.871	0.635	0.724
7	NN-LayerNorm	0.693 $\pm$ 0.051	0.869	0.680	0.772
8	NN-Bottleneck	0.691 $\pm$ 0.034	0.866	0.665	0.761
9	NN-LayerNorm	0.710 $\pm$ 0.044	0.864	0.690	0.778
10	NN-LayerNorm	0.716 $\pm$ 0.038	0.876	0.620	0.700
11	NN-LayerNorm	0.739 $\pm$ 0.036	0.893	0.685	0.763
12	NN-Bottleneck	0.750 $\pm$ 0.030	0.898	0.670	0.748
13	NN-Bottleneck	0.733 $\pm$ 0.037	0.891	0.680	0.762
14	NN-Bottleneck	0.739 $\pm$ 0.052	0.884	0.665	0.744
15	NN-Bottleneck	0.756 $\pm$ 0.057	0.902	0.715	0.784
16	NN-LayerNorm	0.737 $\pm$ 0.044	0.887	0.720	0.805
17	NN-Bottleneck	0.741 $\pm$ 0.061	0.897	0.675	0.754
18	NN-Bottleneck	0.723 $\pm$ 0.055	0.879	0.645	0.731
19	NN-Bottleneck	0.730 $\pm$ 0.044	0.891	0.640	0.727
20	NN-Bottleneck	0.705 $\pm$ 0.038	0.874	0.660	0.756
21	NN-Bottleneck	0.706 $\pm$ 0.026	0.870	0.675	0.760
22	NN-Bottleneck	0.700 $\pm$ 0.043	0.871	0.550	0.606
23	ExtraTrees	0.716 $\pm$ 0.032	0.872	0.745	0.852

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Table 9 – continued from previous page

Layer	Best Probe	AUC (mean $\pm$ std)	AP	Acc	F1
24	NN-Bottleneck	0.723 ± 0.066	0.878	0.585	0.656
25	NN-LayerNorm	0.700 ± 0.049	0.872	0.685	0.778
26	NN-Bottleneck	0.700 ± 0.031	0.863	0.645	0.730
27	NN-Bottleneck	0.710 ± 0.037	0.875	0.669	0.757
28	RBF-SVM	0.696 ± 0.022	0.871	0.735	0.847
29	NN-Bottleneck	0.716 ± 0.058	0.877	0.675	0.754
30	NN-Bottleneck	0.712 ± 0.036	0.883	0.639	0.716
31	NN-Bottleneck	0.731 ± 0.035	0.873	0.629	0.698
32	NN-LayerNorm	0.708 ± 0.033	0.874	0.675	0.765

C Geometric Characterization

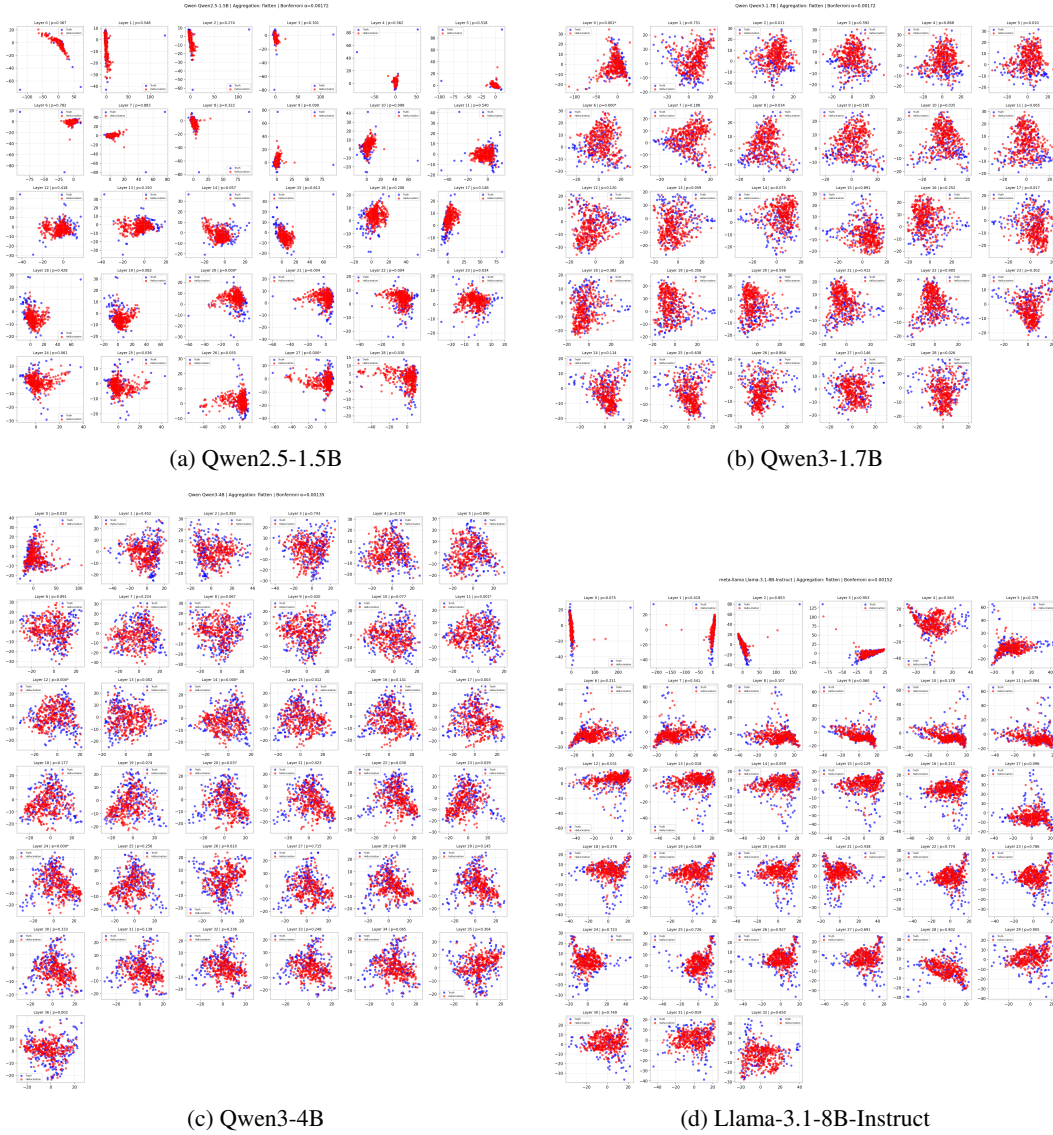


Figure 8: Flattened hidden-state geometry plots for all models. Each panel shows the geometric structure of truthful and hallucinated states under the corresponding model-specific flatten projection.

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